

YOLO Dataset Configuration and Meta Files Report

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Course: CSE (AIML)

Topic: YOLO Dataset Configuration and Meta Files

Introduction

Object detection and segmentation using YOLO models require properly structured datasets and configuration files. These meta files define how the model reads images, labels, and class information.

Dataset YAML File

The YAML file is the main configuration file in YOLO.

It contains:

- Path to dataset
- Training and validation directories
- Class names

Example: path: dataset train: images/train val: images/val

names: 0: person 1: car 2: dog

This file tells the model where data is stored and what objects to detect.

Label Files

Each image has a corresponding `.txt` file.

Format: class_id x_center y_center width height

- class_id → object category
- coordinates → normalized (0 to 1)

These files help the model understand object positions.

Directory Structure

A YOLO dataset is organized as:

dataset/ ├── images/ | ├── train/ | └── val/ ├── labels/ | ├── train/ | └── val/ └── data.yaml

Each image must have a matching label file.

Importance of Meta Files

- Define dataset structure
- Help in training and validation

- Map class IDs to object names
- Ensure correct model predictions

References

- Ultralytics YOLO Documentation
- <https://docs.ultralytics.com>
- <https://docs.ultralytics.com/datasets>

Conclusion

YOLO meta files such as YAML and label files are essential for organizing datasets and enabling accurate object detection and segmentation.